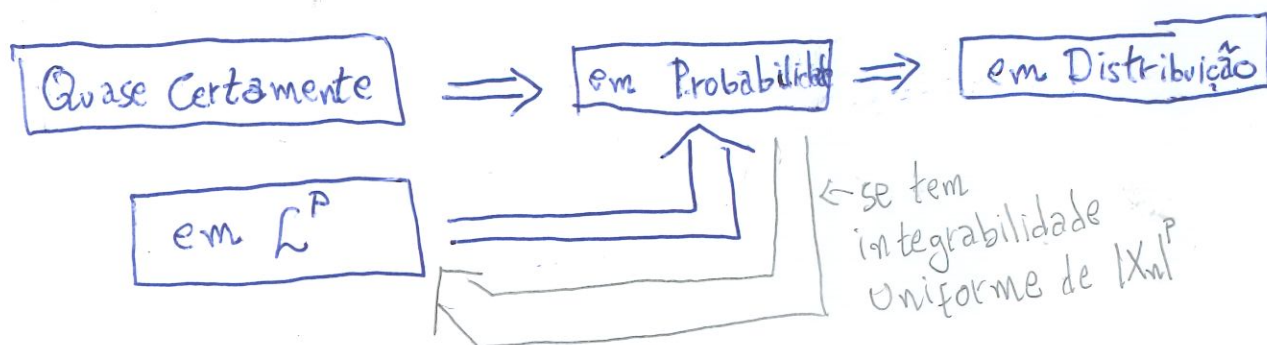


3 Integrabilidade Uniforme



Lem: $X \in L^1 \iff \lim_{a \rightarrow \infty} \mathbb{E}(|X| \mathbb{1}_{\{|X| > a\}}) = 0$

Def: $\{X_\lambda : \lambda \in \Lambda\}$ é uniformemente integrável se

$$\lim_{a \rightarrow \infty} \sup_{\lambda \in \Lambda} \mathbb{E}(|X_\lambda| \mathbb{1}_{\{|X_\lambda| > a\}}) = 0$$

Teo: ^{suponha} $X_n \xrightarrow{P} X$. São equivalentes:

- (i) $(X_n)_{n \in \mathbb{N}}$ é u.i.
- (ii) $X_n, X \in L^1$ e $X_n \xrightarrow{L^1} X$
- (iii) $X_n \in L^1$ e $\mathbb{E}|X_n| \rightarrow \mathbb{E}|X|$

Exemplo: São u.i.:

- (i) $|X_\lambda| \leq Y$, onde $Y \in L^1$ ~~segue de $|X_\lambda| \mathbb{1}_{\{|X_\lambda| > a\}} \leq Y \mathbb{1}_{\{|Y| > a\}}$~~
segue de $|X_\lambda| \mathbb{1}_{\{|X_\lambda| > a\}} \leq Y \mathbb{1}_{\{|Y| > a\}}$
- (ii) $\exists \gamma > 0, K > 0: \mathbb{E}(|X_\lambda|^{1+\gamma}) \leq K \forall \lambda$.
- (iii) $\exists \varphi: \lim_{x \rightarrow \infty} \frac{\varphi(x)}{x} = +\infty$ e $\mathbb{E}(\varphi(|X_\lambda|)) \leq K \forall \lambda$.

② Exemplo: $(X_n)_{n \in \mathbb{N}}$ iid $\mathbb{P}(X_n=0) = \mathbb{P}(X_n=2) = 1/2$

- $M_n = \prod_{k=1}^n X_k$

- $(M_n)_{n \in \mathbb{N}}$ é martingal

- $\mathbb{E}|M_n| = 1$

$\Rightarrow M_n \rightarrow M_\infty$ q.c.

- $M_\infty = 0$

- $\mathbb{E}(M_\infty) = 0 \neq 1 = \mathbb{E}(M_n)$

$\Rightarrow M_n \not\rightarrow_{L^1} M_\infty$

① Processos Estocásticos (Dia 30/09/2026)

Ultima aula: $(M_n)_{n \in \mathbb{N}}$ martingal, $\mathbb{E}(M_n^2) \leq C \quad \forall n \in \mathbb{N}$
 $\Rightarrow \exists M_\infty$ t.q. $\mathbb{E}(M_\infty^2) \leq C$ e $M_n \rightarrow M_\infty$ q.c e em L^2

Prova: provamos $\exists M_\infty : M_n \rightarrow M_\infty$ q.c

- $d_k = M_k - M_{k-1}$

- $\mathbb{E}(d_k d_j) = 0 \quad \forall j \neq k$

- $C \geq \mathbb{E}(M_n^2) = \sum_{k=0}^n \mathbb{E}(d_k^2) \Rightarrow C \geq \sum_{k=0}^{\infty} \|d_k\|_2^2$

$\Rightarrow \underbrace{\left(\sum_{k=0}^n d_k \right)}_{M_n} \text{ é convergente em } L^2(\Omega, \mathcal{F}, \mathbb{P})$
(porque $L^2(\Omega, \mathcal{F}, \mathbb{P})$ é completo)
(e $(M_n)_{n \in \mathbb{N}}$ é Cauchy)

$\exists \tilde{M}_\infty : M_n \rightarrow \tilde{M}_\infty$ em L^2 ~~pois~~

$\Rightarrow M_n \rightarrow \tilde{M}_\infty$ em $\mathbb{P}$ e $M_n \rightarrow M_\infty$ em $\mathbb{P}$

$\Rightarrow \tilde{M}_\infty = M_\infty$ q.c

$\Rightarrow M_n \rightarrow M_\infty$ q.c e em L^2 (em particular $\|M_n\|_2 = \lim_{n \rightarrow \infty} \|M_n\|_2$)